Date Q1 Quiz 2 - (A)

ⓐ $f: \forall x \forall y (x+y = y+x)$

$\sim f: \exists x \exists y$ such that $x+y \neq y+x$

There exists atleast one pair of values x and y s.t $x+y$ is not equal to $y+x$.

ⓑ $f: \exists y \forall x (2x^2+1 > x^2 y)$

$\sim f: \forall y \exists x$ such that $2x^2+1 \leq x^2 y$

For every possible y, there exists atleast one x s.t $2x^2+1$ is less than or equal to $x^2 y$

ⓒ $f: \forall x \forall y \exists z$ such that $2z = x+y$

$\sim f: \exists x \exists y \forall z (2z \neq x+y)$

There exist some values x and y s.t no value of z will satisfy $2z = x+y$

Q2

ⓐ All math problems are either easy or unsolvable.

ⓑ A math problem is difficult if and only if it is not unsolvable.

Q3

ⓐ $\forall x (P(x) \rightarrow Q(x))$

ⓑ $\exists x (R(x) \wedge \sim Q(x))$

ⓒ $\exists x (R(x) \wedge \sim P(x))$

ⓓ Yes; from ⓑ there exists a 'c' excuse s.t

$R(c) \wedge \sim Q(c)$ ∴ existential instantiation

↳ c is an excuse that is unsatisfactory

i.e. $\sim Q(c)$ is true

$P(c) \rightarrow Q(c)$

$\sim Q(c)$

$\therefore \sim P(c)$ ∴ modus tollens

Hence, $\therefore R(c) \wedge \sim P(c)$